FULL-CORPUS BENCHMARK / TORGO

Recognizing male dysarthric speech

Six downloaded models | 1,920 recordings | 5 speakers
Prepared October 20, 2026 at 01:36 PM EDT

Parakeet TDT v3 has the lowest equal-weight speaker WER in this run: **60.1%**. Its corpus WER is **57.8%**. Lower WER/CER is better; higher exact-match rate is better.

Rank / model	Speaker mean WER	Corpus WER	Corpus CER	Exact match
1. Parakeet TDT v3	60.1%	57.8%	42.4%	22.7%
2. Vosk Large	82.7%	82.1%	52.8%	27.7%
3. Wav2Vec2 Base	86.1%	84.3%	48.2%	21.8%
4. Vosk Lgraph	88.3%	87.7%	54.8%	26.1%
5. Vosk Small	94.8%	94.0%	60.1%	22.8%
6. Whisper Small	388.0%	367.5%	523.6%	29.3%

Coverage: 11,520 model-recording attempts; 18 runtime failures; 1,902 recordings completed by every model.

Primary ranking: calculate corpus WER within each speaker, then average the five speaker WERs equally. This prevents speakers with more recordings from dominating the primary score.

Task trade-off: Vosk Large has the highest single-word exact-match rate (31.4%). Whisper Small has the highest whole-clip exact-match rate (29.3%), but its extra and repetitive output on other clips drives substantially higher WER. The primary ranking measures error burden, not just the number of perfectly matched clips.

Interpretation: this is agreement with TORGO's supplied prompts on the selected head-microphone cohort. It is not clinical validation, a guarantee for a new speaker, or a measure of meaning understood. No model was tuned or fine-tuned for this run.

WER can exceed 100% when extra words are predicted. It must not be converted into a bounded accuracy percentage by subtracting it from 100%.

What was evaluated

All accepted recordings in the existing TORGO head-microphone manifest were evaluated once per model.

Speaker	Recordings	Audio (min)	Reference words
M01	369	24.2	925
M02	385	24.8	949
M03	401	19.4	1011
M04	272	21.5	698
M05	493	36.7	1245

Total distinct audio: **2.11 hours**; approximately **4,828 reference words**. Speaker IDs M01-M05 are the male dysarthric cohort; controls are excluded.

Import exclusions

Exclusion / session issue	Count
annotated or phonetic prompt requires review	75
image prompt requires a verbatim transcript	23
missing prompt	1
prompt marked xxx (unusable)	18
missing wav_headMic	1

Only head-microphone audio was selected. M04/Session1 has no head-microphone folder and was not substituted with array-microphone audio. Two microphone recordings of the same utterance were not double-counted.

Procedure and fairness

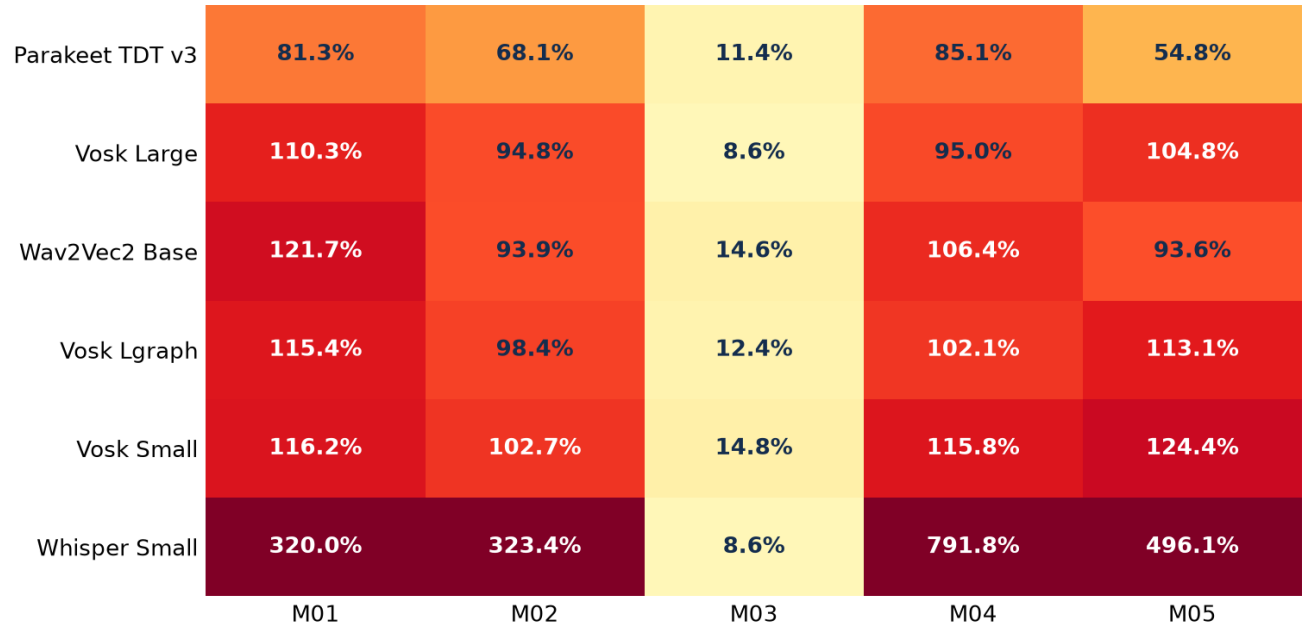
Every model received the same decoded 16 kHz, mono, signed 16-bit PCM per sample. PCM SHA-256 values were checked across successful model outputs. All models attempted every manifest ID. Rankings use only recordings successfully completed by all six models, with failures and coverage reported separately.

Independent QA: RapidFuzz 3.14.6 verified 23,004 word/character edit totals with zero mismatches. All 1,920 original PCM files were rehashed and matched the saved inputs.

This corpus includes single-word and multi-word stimuli. It contains no dedicated silence benchmark in this manifest. Prompt matching and basic exclusions were automated; references have not been exhaustively verified against the speech.

Performance by speaker

Corpus WER within each speaker. The same common-success recordings are used for every model.



Color intensity is capped at 150% WER; cell labels show the full, unbounded scores.

Speaker	Lowest WER model	WER	Common clips
M01	Parakeet TDT v3	81.3%	369
M02	Parakeet TDT v3	68.1%	385
M03	Vosk Large / Whisper Small	8.6%	401
M04	Parakeet TDT v3	85.1%	272
M05	Parakeet TDT v3	54.8%	475

Use these rows when choosing a model for a particular individual. An overall winner need not win for every speaker. Do not infer impairment severity from recognition errors: severity labels were not independently assessed here.

Words, longer utterances, and averages

Model	Word clips	Word exact match	Multi-word clips	Multi-word WER	Multi-word exact
Parakeet TDT v3	1444	21.8%	458	43.7%	25.3%
Vosk Large	1444	31.4%	458	75.8%	15.7%
Wav2Vec2 Base	1444	24.4%	458	79.6%	13.3%
Vosk Lgraph	1444	30.3%	458	81.7%	13.1%
Vosk Small	1444	26.3%	458	89.6%	11.8%
Whisper Small	1444	29.6%	458	223.0%	28.6%

A single-word clip has one whitespace-delimited token after normalization; multi-word means two or more. Multi-word clips include sentences and passage fragments. These groups are derived from reference text, not manually validated task labels.

Three ways to average errors

Model	Corpus WER	Mean speaker WER	Mean clip WER	Median clip WER
Parakeet TDT v3	57.8%	60.1%	79.5%	100.0%
Vosk Large	82.1%	82.7%	91.8%	100.0%
Wav2Vec2 Base	84.3%	86.1%	91.8%	100.0%
Vosk Lgraph	87.7%	88.3%	97.0%	100.0%
Vosk Small	94.0%	94.8%	101.1%	100.0%
Whisper Small	367.5%	388.0%	588.0%	100.0%

Corpus WER: total word edits divided by total reference words. Longer references receive more weight.
Mean speaker WER: the mean of speaker-level corpus WERs. Each speaker receives equal weight.
Mean clip WER: the mean of per-recording WERs. A one-word recording and a long sentence receive equal weight.
Exact match: normalized prediction equals normalized reference in full. This is a bounded percentage of complete matches.

The corpus and speaker-level averages answer different questions. Rank by the speaker mean for broad cohort balance, then inspect the individual and task-length results before choosing a model.

How the models fail

Word edit components below are divided by the common subset's reference word count. S + D + I equals corpus WER.

Model	Substitutions	Deletions	Insertions	Total WER
Parakeet TDT v3	35.3%	13.4%	9.1%	57.8%
Vosk Large	58.0%	2.4%	21.6%	82.1%
Wav2Vec2 Base	63.8%	2.6%	17.9%	84.3%
Vosk Lgraph	59.7%	1.8%	26.1%	87.7%
Vosk Small	64.4%	2.2%	27.4%	94.0%
Whisper Small	45.8%	3.0%	318.7%	367.5%

A substitution replaces a reference word; a deletion omits it; an insertion adds a word. A model that produces fluent extra text can have substantially worse WER than a model that returns only a few words.

Model	Empty outputs	Expansion flags	Non-Latin flags
Parakeet TDT v3	369	1	386
Vosk Large	15	2	0
Wav2Vec2 Base	17	1	0
Vosk Lgraph	15	1	0
Vosk Small	23	3	0
Whisper Small	1	105	158

Empty output: the normalized prediction has no text.
Expansion flag: predicted word count is at least three times the reference count and at least ten words.
Non-Latin flag: at least one alphabetic character has a non-Latin Unicode name.

These are review flags, not proven hallucinations or automatic failure labels. They may overlap and do not measure factual correctness. Raw predictions are preserved in samples.csv, including non-Latin and repeated output.

The multilingual models produced non-Latin flags on 386 Parakeet outputs and 158 Whisper outputs. Whisper used automatic language detection, and Parakeet also selects output language automatically. A separate language-controlled experiment could help distinguish language-selection failures from recognition failures; it was not substituted into this baseline.

Runtime and reliability

Timing is descriptive: model workers ran concurrently on this computer, not under isolated laboratory conditions.

Model	Inference min	Median sec	P95 sec	Overall RTF	P95 RTF
Parakeet TDT v3	37.5	0.86	2.99	0.30	1.28
Vosk Large	24.7	0.35	3.26	0.20	0.38
Wav2Vec2 Base	4.3	0.11	0.27	0.03	0.05
Vosk Lgraph	153.7	3.79	11.51	1.21	2.22
Vosk Small	17.4	0.45	1.23	0.14	0.25
Whisper Small	651.4	7.08	155.69	5.15	42.63

RTF = inference seconds / recording seconds. An overall RTF below 1 means aggregate processing was faster than audio playback in this run; it does not establish streaming response latency. P95 is the 95th percentile across successful recordings. Inference totals across concurrent workers must not be summed and described as elapsed wall-clock time. Timing statistics use successful attempts only; unfinished calls interrupted during checkpoint rescheduling are excluded.

Model	Completed / attempted	Coverage	First load sec
Parakeet TDT v3	1,920 / 1,920	100.0%	4.29
Vosk Large	1,920 / 1,920	100.0%	3.47
Wav2Vec2 Base	1,902 / 1,920	99.1%	3.93
Vosk Lgraph	1,920 / 1,920	100.0%	0.46
Vosk Small	1,920 / 1,920	100.0%	0.17
Whisper Small	1,920 / 1,920	100.0%	3.98

Recorded runtime failures: Wav2Vec2 Base: 18. Most common error (18 cases): RuntimeError: Calculated padded input size per channel: (2). Kernel size: (3). Kernel size can't be greater than actual input size. Exact sample IDs and all error messages are retained in analysis.json.

What timing includes

The timed region is adapter.transcribe(): waveform preparation inside the adapter, feature extraction, decoding, and transcript production. WAV decoding, score calculation, checkpoint writes, and initial weight loading are excluded. Vosk recognizer creation is included per recording. Load time is reported separately.

Environment: macOS-27.0.1-arm64-arm-64bit; 14 logical CPUs; CPU inference. Default PyTorch intra-op threads: 4; inter-op threads: 1. Whisper and Parakeet partition workers used one intra-op thread each; Vosk Lgraph used four independent recognizers. Processor: Apple M4 Pro; 48 GiB RAM. Transformer adapters use float32. No explicit warm-up or repeat timing trials were performed.

Python 3.11.17; Vosk 0.3.44; Transformers 5.20.2; PyTorch 2.14.1.

Workers, brief diagnostic runs, and report preparation shared system resources. The first Vosk Small checkpoint rows were measured before worker concurrency began. Timing differences are not a controlled speed ranking. Load times show the first saved process initialization, not the sum across partition workers. Use a repeated, isolated deployment benchmark for a speed-critical decision.

Example predictions

Two deterministically selected examples illustrate variation. They are not a representative accuracy sample.

M01/Session1/headMic/0006

Supplied reference: trait

Model	Predicted text	WER
Parakeet TDT v3	[empty output]	100.0%
Vosk Large	grey	100.0%
Wav2Vec2 Base	TRADE	100.0%
Vosk Lgraph	grey	100.0%
Vosk Small	hey	100.0%
Whisper Small	CREATE!	100.0%

M01/Session1/headMic/0009

Supplied reference: Grandfather likes to be modern in his language.

Model	Predicted text	WER
Parakeet TDT v3	Grandpa in his language.	62.5%
Vosk Large	ruin the the doobie i do need leaving or	112.5%
Wav2Vec2 Base	DRAN VALY LEG BOBY MY DO MA MA THE ANGUT	125.0%
Vosk Lgraph	ruin the they do be i do the leaving or	112.5%
Vosk Small	when eg will be my do mean the viewing angle	112.5%
Whisper Small	[Non-Latin text; complete prediction is in samples.csv]	612.5%

Examples are the first eligible single-word and 3-10-word common-success clips by sorted sample ID. Long predictions are shortened for display; predictions containing non-Latin scripts are replaced with a label in this PDF only. Complete text and scores remain in the CSV and JSON outputs.

Data-quality sensitivity check

The benchmark contains **1 recording(s) longer than 30 seconds**. All are retained in the primary scores. This secondary diagnostic removes only those IDs from the common-success subset; no predictions or reference words are edited. A further **30 recordings are shorter than 100 milliseconds**; several are paired with full-sentence prompts. These are flagged for audio/reference review, not silently removed.

Recording	Seconds	Supplied reference
M01/Session2_3/headMic/0180	179.94	hill

Model	Full corpus WER	Remove >30s	Remove all duration flags
Parakeet TDT v3	57.8%	56.2%	55.9%
Vosk Large	82.1%	79.7%	79.6%
Wav2Vec2 Base	84.3%	81.8%	81.8%
Vosk Lgraph	87.7%	85.1%	85.0%
Vosk Small	94.0%	91.6%	91.6%
Whisper Small	367.5%	364.2%	363.8%

Comparable clips: full 1,902; without long clips 1,901; without either duration flag 1,889.

The 179.94-second M01 recording is paired with the one-word prompt 'hill', while the next-longest accepted recording is about 23 seconds. Duration alone does not prove the recording is invalid. Listen to this file and verify prompt alignment before using a cleaned dataset for a separate evaluation.

Short-file example	Millisec	Supplied reference
M05/Session2/headMic/0303	42.9	He took me out one day.
M05/Session2/headMic/0330	9.4	Then we'd really have someplace to go.
M05/Session2/headMic/0331	59.7	walk

All short-file IDs are listed in analysis.json. Clips that cause any model to fail are already excluded from every model's common-success accuracy scores; they remain in coverage and failure counts.

Why this matters

A long recording paired with a very short reference can accumulate many insertion errors. Word error rates are unbounded, so a single problematic clip can noticeably affect pooled and speaker-level scores. The full-corpus ranking remains the primary result; the diagnostic is clearly separated to avoid silently discarding a difficult example after observing model behavior.

To improve the next benchmark, review reference/audio alignment and recording notes, correct references independently of model predictions, and freeze a new manifest before rerunning every model. Keep original and cleaned results as separate experiments.

How to use these results

First candidate for further validation: Parakeet TDT v3, based on the lowest equal-weight speaker WER in this fixed cohort. Inspect speaker-level results rather than assuming the same model will be best for a new individual. Test the chosen configuration on fresh recordings from the intended user before selecting it for deployment.

For isolated words: start by comparing Vosk Large, the single-word exact-match leader at 31.4%, with the speaker-specific alternatives. The overall WER leader and the word-match leader answer different use cases. Substantial errors remain in every evaluated configuration.

Limits that affect the conclusion

1. **References:** TORGO supplies intended prompts. Omissions, restarts, interviewer speech, or deviations from the prompt can affect scores. The references were not exhaustively manually corrected.
2. **Cohort:** five male speakers, one microphone type, and the imported sessions. No female, control, new-speaker, background-noise, or silence generalization claim follows from this run.
3. **Normalization:** Unicode NFKC, case folding, punctuation-to-space replacement, and whitespace collapse; apostrophes are preserved. Written numbers are not converted to spoken-number forms; contractions are not expanded. CER includes spaces. Thus correct-looking alternate text formatting can still incur errors.
4. **Task:** final transcripts were compared, not semantic meaning, speaker attribution, streaming stability, word-timestamp accuracy, or lipsync alignment.
5. **Inference:** one fixed decoding configuration per model; no language-model tuning, fine-tuning, or hyperparameter search. Public-corpus overlap with pretraining data is unknown. No statistical significance or population-level confidence interval is claimed from five speakers.

Decoding configurations

Vosk Small: en-us-0.15; Vosk Lgraph: en-us-0.22-lgraph; Vosk Large: en-us-0.22. All use default unrestricted recognizers and a fresh recognizer per clip. Wav2Vec2: facebook/wav2vec2-base-960h, greedy CTC without an external language model, 30-second chunks with 5-second stride. Parakeet: nvidia/parakeet-tdt-0.6b-v3, default generation over each recording. Whisper: openai/whisper-small, transcription task with automatic language detection, timestamp-enabled generation, five beams, and max_length=448 from the loaded model/pipeline configuration.

Reproducibility and supporting files

Manifest SHA-256: 20d17e090714aeedd070365373e86ca04086b12de9a1c0aa664d0ca5e208bc04
Raw report SHA-256: 6263cb2e071f900d045b6b38df09d724d443527ff93a910016cfd6c841749818

Run directory: results/torgo-full-2026-10-19
report.json and samples.csv preserve all predictions; analysis.json, leaderboard.csv, and per-speaker.csv contain calculated summaries. Per-model JSONL checkpoints, metadata JSON, logs, run-config.json, and execution.json document completion and environment details. Rebuild this report with scripts/make_benchmark_report.py and the same raw report and manifest.

Dataset documentation and required publication citation: University of Toronto TORGO database. Rudzicz, Namasivayam and Wolff (2012), The TORGO database of acoustic and articulatory speech from speakers with dysarthria, Language Resources and Evaluation 46(4), 523-541. Academic-use terms apply.

Model specifications and Raspberry Pi fit

Reference device: **Raspberry Pi 5, 8 GB RAM**, 64-bit OS, active cooling, CPU inference, one model at a time. Pi assessments below are engineering judgments, not measured Pi results.

Downloaded model	Parameters	Local files	Memory guidance	Pi 5 / 8 GB assessment
Vosk Small en-us-0.15 [1]	Not published	68 MiB	About **300 MB runtime** is Vosk's typical small-model guidance; not a measurement of this run.	**Strongest starting point.** Officially targets Raspberry Pi. Validate latency with the lipsync app running.
Vosk Lgraph en-us-0.22-lgraph [1]	Not published	204 MiB	Model-specific runtime RAM is **not published**. Dynamic decoding graph; storage size does not predict peak RAM.	**Candidate to test.** Compact files make it worth a Pi trial; measure peak RAM and latency before committing.
Vosk Large en-us-0.22 [1]	Not published	2.67 GiB	Vosk's big-model category can use **up to 16 GB RAM**. This is not a measured requirement for en-us-0.22.	**Poor default choice.** 8 GB may be insufficient. Even a 16 GB Pi needs a memory/latency trial and OS headroom.
Whisper Small openai/whisper-small [2]	244 M	926 MiB	FP32 weights alone: **~0.98 GB**. Total runtime RAM is higher. Alternate whisper.cpp reports ~852 MB for small [5].	**Conditional.** Try whisper.cpp, which supports Pi and quantization. Current FP32, five-beam Python decoding needs testing.
Wav2Vec2 Base wav2vec2-base-960h [3]	94.4 M	360 MiB	FP32 weights alone: **~0.38 GB**. Framework and audio activations add RAM; no total-RAM figure is published on the card.	**Plausible candidate.** Smallest neural model here; keep chunking. ARM dependency support and speed must be checked.
Parakeet TDT v3 0.6b-v3 [4]	~600 M	2.34 GiB	FP32 weights alone: **~2.4 GB**. Card says at least 2 GB to load (configuration dependent); total FP32 RAM is higher.	**Experimental.** Loading short clips may fit, but highest neural weight cost here. Optimized for NVIDIA GPUs; Pi has no CUDA.

How to read the memory figures

Peak RAM was not measured in this benchmark. FP32 figures are calculated as published parameter count x 4 bytes, rounded; they exclude Python/PyTorch, activations, decoder caches, audio buffers, loading copies, and the OS. Longer audio, beams, or concurrent models can raise memory use. They are lower bounds, not safe total-RAM budgets. The whisper.cpp figure uses a different runtime/representation from this benchmark.

Local files are the rounded sum of file bytes in each downloaded directory, inspected October 20, 2026 (MiB/GiB are binary; MB/GB are decimal). They are neither ZIP sizes nor runtime RAM. Vosk's cards do not give parameter counts; graph and language-model files also consume storage and memory.

Deployment decision

Start with Vosk Small for a Pi prototype, then compare Lgraph and Wav2Vec2 on the target speaker. Parakeet leads this cohort's WER, but needs a separate Pi feasibility trial. Loading successfully does not establish real-time performance. Measure peak resident memory, sustained latency, and thermal behavior on the actual board. Recheck WER after changing runtime, quantization, chunking, or decoding.

Sources (accessed October 20, 2026)

[1] Vosk model catalog / runtime guidance; [2] OpenAI Whisper Small; [3] Meta Wav2Vec2 Base 960h; [4] NVIDIA Parakeet TDT 0.6b v3; [5] whisper.cpp memory / Pi support; [6] Raspberry Pi 5 specifications (quad-core 2.4 GHz Arm Cortex-A76, RAM options up to 16 GB).